# Scale and Similarity

## Concept

Similar shapes have equal matching angles and proportional matching lengths.

### Scale factor

k = image length/original length.

### Scale drawing

1:200 means 1 unit on paper represents 200 matching units.

### Match sides

Use corresponding sides in the same order in every ratio.

### Metric units

1 m = 100 cm; convert before comparing lengths.

## Example 1

Sides 6, 8, 10 are enlarged so the side 6 becomes 9.

### Solution

1. Scale factor = 9/6 = 1.5.

2. Other sides: 8(1.5) = 12 and 10(1.5) = 15.

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## Example 2

A 1.5 m stick casts a 2 m shadow. A tree casts a 12 m shadow.

### Solution

1. Assume level ground and the same sun angle: $h/12 = 1.5/2$.

2. $h = 12(1.5/2) = 9$ m.

# Scale and Similarity

## Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Enlarge sides 4, 7, 9 by scale factor 2.5.

2. On a 1:50 plan, what actual distance does 8 cm represent?

3. Corresponding sides are 12 and 18. If another original side is 10, find the corresponding enlarged side.

4. Two rectangles are 4 by 6 and 6 by 8. Are they similar?

# Scale and Similarity

## Practice answers | check after trying

1. 10, 17.5 and 22.5.

2. $8 \times 50 = 400$ cm $= 4$ m.

3. $k = 18/12 = 1.5$, so the new side is 15.

4. No: $6/4 = 1.5$ but $8/6 = 4/3$. The corresponding scale factors differ, even though all angles are right angles.